Parallel and Distributed Computing
Lecture 1

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Text Book

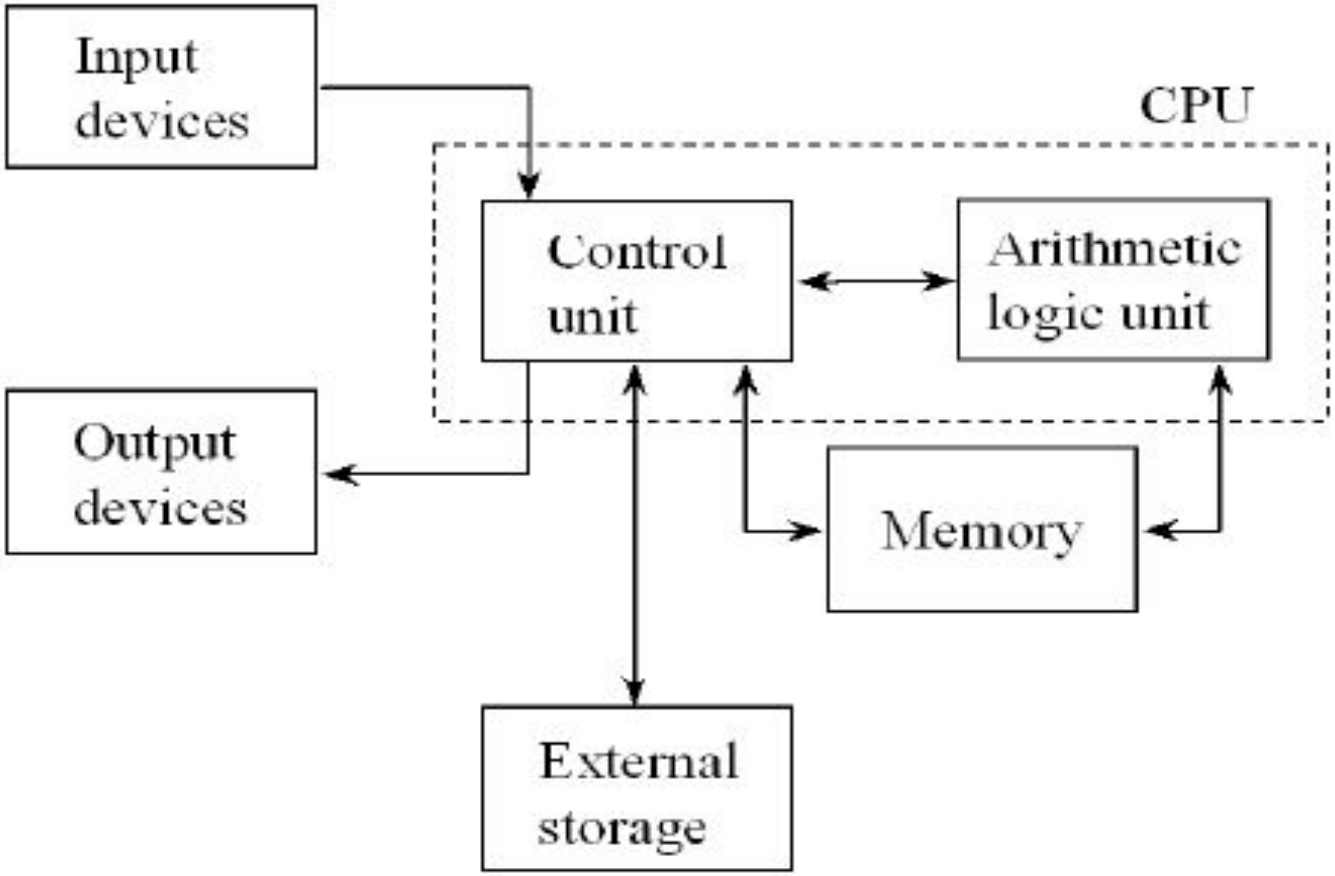
Introduction to Parallel Computing by Ananth Grama and Anshul Gupta

Distributed Systems: Concept and Design by George Coulouris, Gordon Blair

Using OpenMP: Portable Shared Memory Parallel Programming by Barbara Chapman, Gabriele Jost, Ruud van der Pas

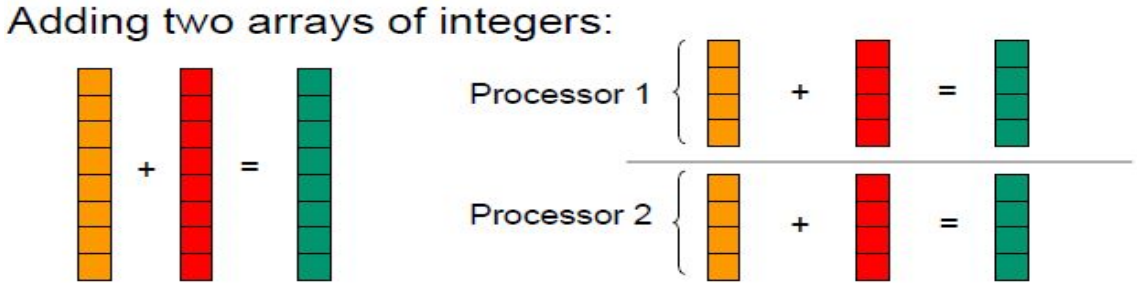
Lecture 1: Introduction

von Neumann Architecture



Parallel and Distributed Computing

- Parallel computing (processing):
  - the use of two or more processors, ***usually within a single system***, working simultaneously to solve a single problem.
- Distributed computing (processing):
  - any computing that involves ***multiple computers remote from each other*** that each have a role in a computation problem or information processing.
- Parallel programming:
  - the human process of developing programs that express what computations should be executed in parallel.



What is Parallel Computing?

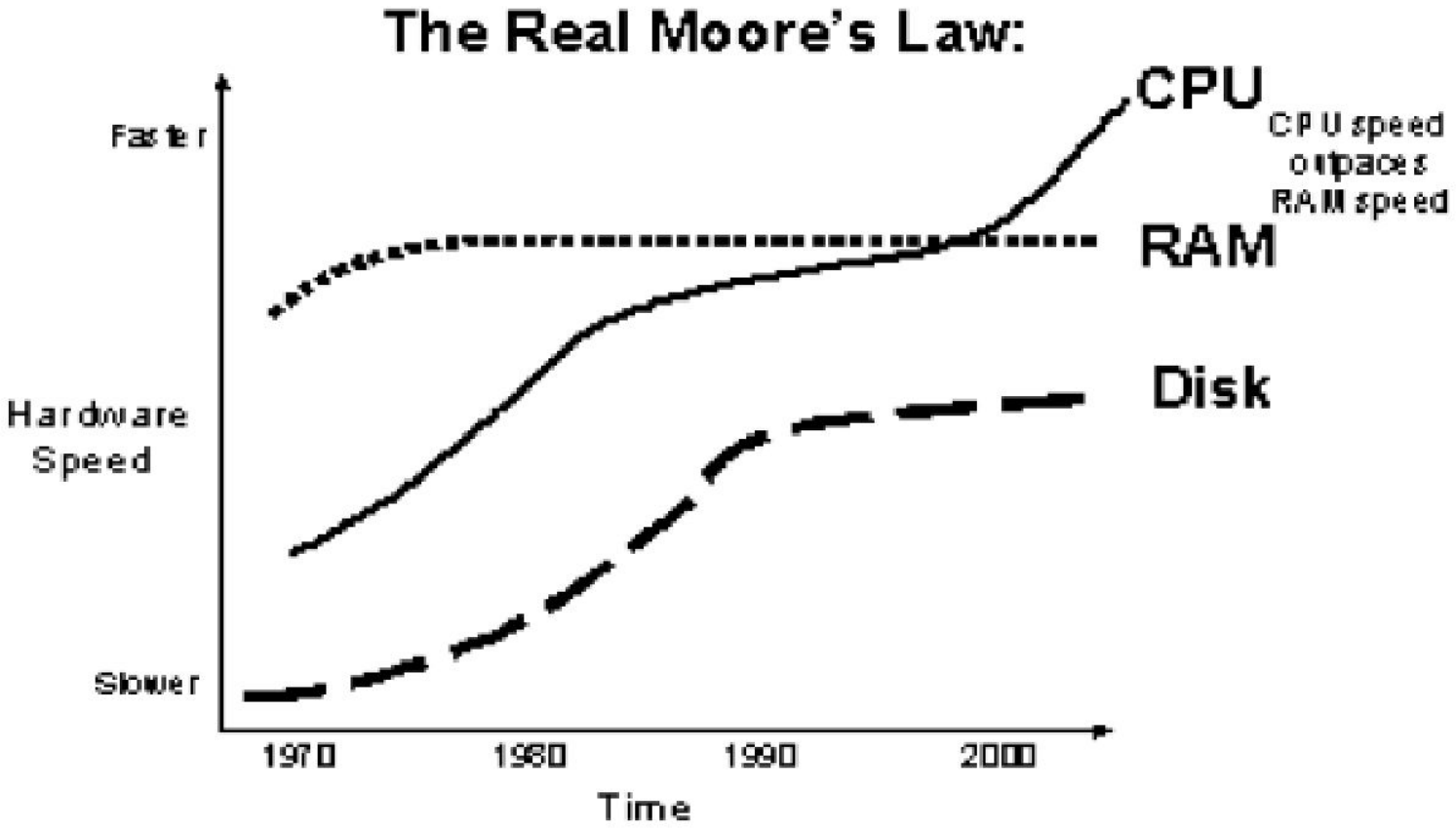
Serial Computing:

- Traditionally, software has been written for ***serial*** computation:
  - A problem is broken into a discrete series of instructions
  - Instructions are executed sequentially one after another
  - Executed on a single processor
  - Only one instruction may execute at any moment in time

CPU and Memory Speeds

- In 20 years, CPU speed (clock rate) has increased by a factor of **1000**
- DRAM speed has increased only by a factor of smaller than **4**
- How to feed data *faster enough* to keep CPU busy?
- CPU speed: 1-2 ns
- DRAM speed: 50-60 ns
- Cache: 10 ns

CPU, Memory, and Disk Speed

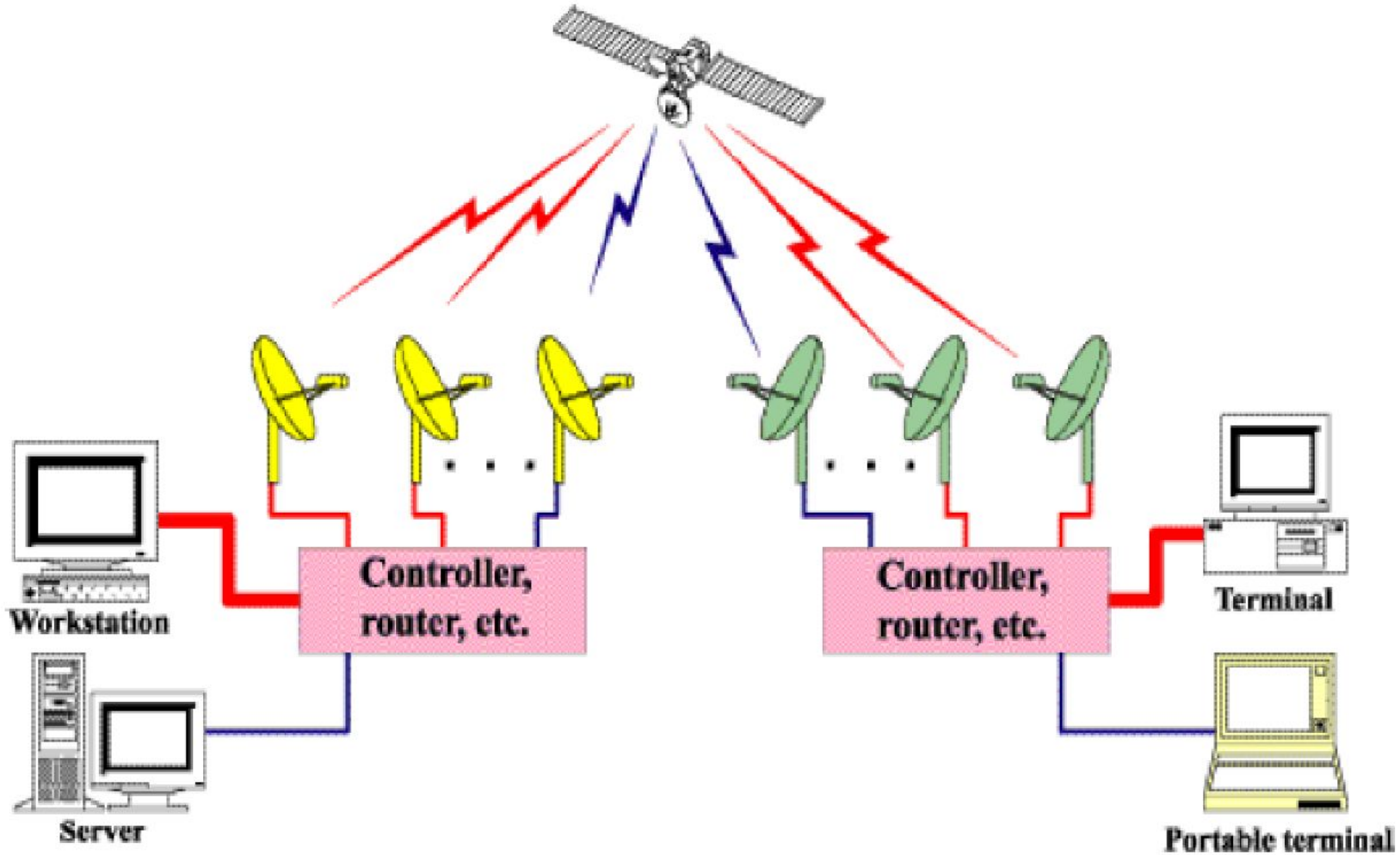


Possible Solutions

Distributed Data Communications

- Data may be collected and stored at different locations
- It is expensive to bring them to a central location for processing
- Many computing assignments many be inherently parallel
- Privacy issues in data mining and other large scale commercial database manipulations

Distributed Data Communications



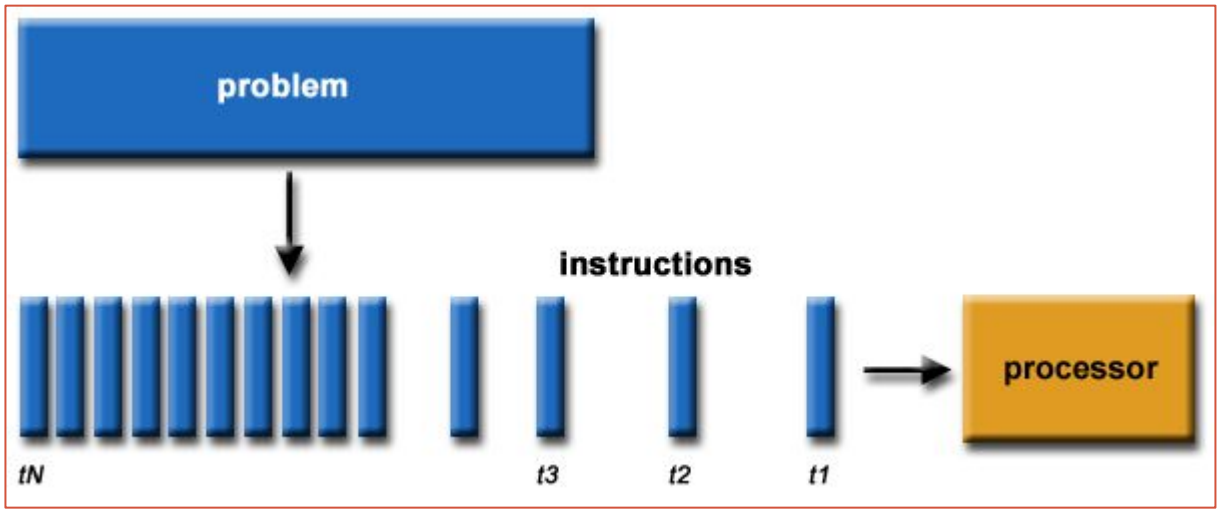
Why Use Parallel Computing

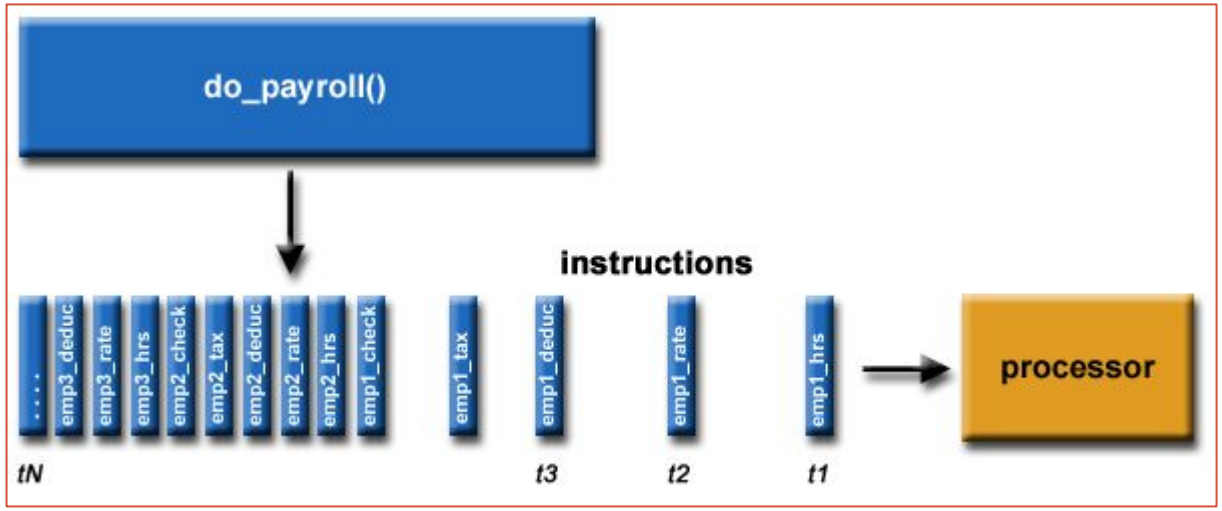
- Save time –many processors work together
- Solve larger problems – larger than one processor's CPU and memory can handle
- Provide concurrency – do multiple things at the same time: online access to databases, search engine
- Google's 4,000 PC servers are one of the largest clusters in the world

Why Parallel Computing?

- To solve larger problems
- – many applications need significantly more memory than a regular PC can provide/handle
- To solve problems faster
- – despite of many advances in computer hardware technology, many applications are running slower and slower
  - e.g. databases having to handle more and more data
  - e.g. large simulations working on even more accurate solutions

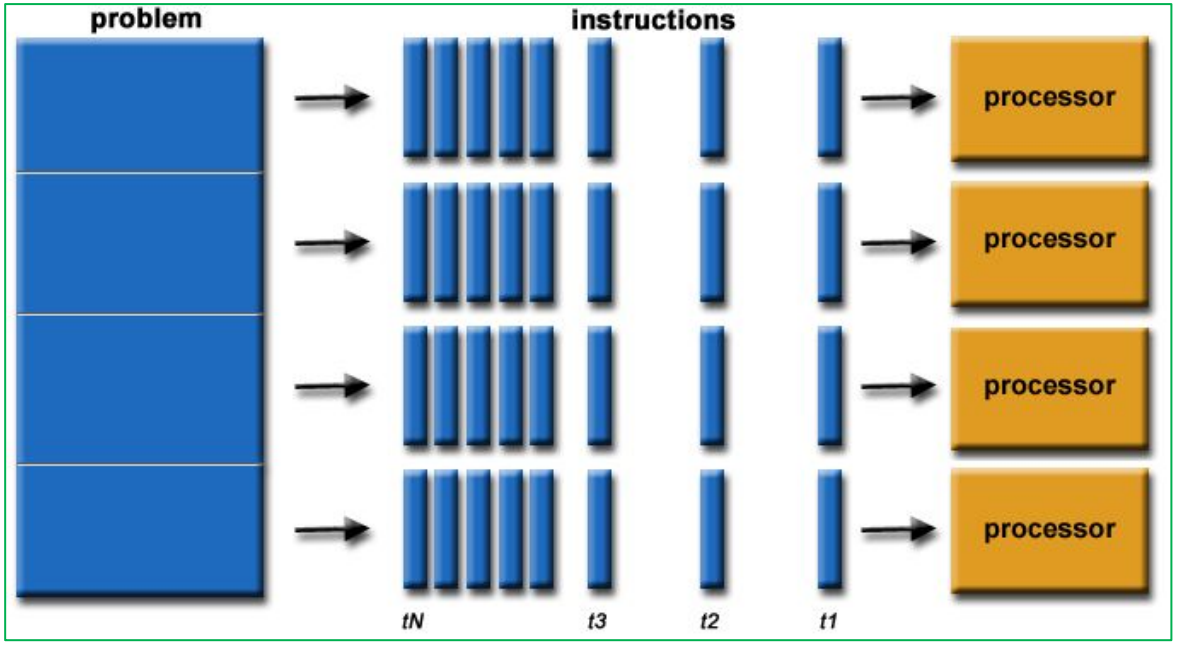
Serial execution

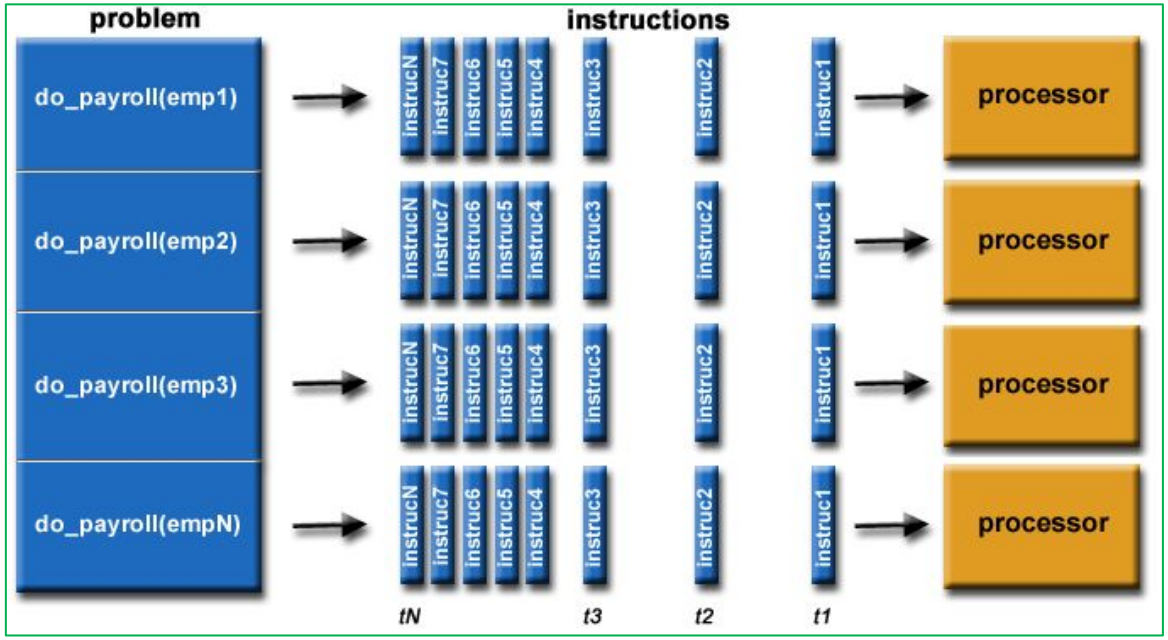




Parallel Computing:

- In the simplest sense, ***parallel computing*** is the simultaneous use of multiple compute resources to solve a computational problem:
  - A problem is broken into discrete parts that can be solved concurrently
  - Each part is further broken down to a series of instructions
  - Instructions from each part execute simultaneously on different processors
  - An overall control/coordination mechanism is employed


problem
instructions
processor
processor
processor
processor
tN
t3
t2
t1


problem
instructions
do_payroll(emp1)
do_payroll(emp2)
do_payroll(emp3)
do_payroll(empN)
instrucN
instruc7
instruc6
instruc5
instruc4
instruc3
instruc2
instruc1
processor
processor
processor
processor
tN
t3
t2
t1

The Inherent Need for Speed

- We want things done ***fast***. If we can get it by the end of the week, we actually want it tomorrow. If we can get it tomorrow, we would really like it today. Let's face it, we're a society that doesn't like to wait.
- Just think about the last time **you stood in line** at a fast food restaurant and had to wait for more than a couple of minutes for your order.
- This idea extends to other things like the **weather.** We routinely check the hourly forecast to see what the weather will be like on our commute to and from work. We expect that there is a computer, behind the scenes, providing this information.

- But did you know that a single computer is often not up to the task? That is where the idea of parallel computing comes in.

Parallel Computing

- In simple terms, **parallel computing** is breaking up a task into smaller pieces and executing those pieces at the same time, each on their own processor or on a set of computers that have been networked together. Let's look at a simple example. Say we have the following equation:
- Y = (4 x 5) + (1 x 6) + (5 x 3)
- On a single processor, the steps needed to calculate a value for Y might look like:
- Step 1: Y = 20 + (1 x 6) + (5 x 3)
- Step 2: Y = 20 + 6 + (5 x 3)
- Step 3: Y = 20 + 6 + 15
- Step 4: Y = 41

- But in a parallel computing scenario, with three processors or computers, the steps look something like:
- Step 1: Y = 20 + 6 + 15
- Step 2: Y = 41
- Now, this is a simple example, but the idea is clear. Break the task down into pieces and execute those pieces simultaneously.

Parallel Vs Distributed Computing

- Parallel computing is a computation type in which multiple processors execute multiple tasks simultaneously.
- Distributed computing is a computation type in which networked computers communicate and coordinate the work through message passing to achieve a common goal.

Number of Computers Required

- Parallel computing occurs on one system.
- Distributed computing occurs between multiple system.

Processing Mechanism

- In parallel computing multiple processors perform processing.
- In distributed computing, computers rely on message passing.

Memory

- In Parallel computing, computers can have shared memory or distributed memory.
- In Distributed computing, each computer has their own memory.

Usage

- Parallel computing is used to increase performance and for scientific computing.
- Distributed computing is used to share resources and to increase scalability.

Synchronization

- All processors share a single master clock for synchronization in parallel computing.
- There is no global clock in distributed computing, it uses synchronization algorithms.

Task 1

The fastest computer in the world today

- What is its name?
- Where is it located?
- How many processors does it have?
- What kind of processors?
- How fast is it?

The fastest computer in the world today

- What is its name? IBM Summit (a supercomputer)
  - Uses: The Summit supercomputer provides scientists and researchers the opportunity to solve complex tasks in the fields of energy, artificial intelligence, human health and other research areas. It has been used in *Earthquake Simulation*, *Extreme Weather simulation* using AI, *Material science*, *Genomics* and in predicting the lifetime of Neutrinos in physics
- Where is it located? United States
- How many processors does it have (architecture)? 4,611,236 processor cores
- What kind of processors? 9216 POWER9 22-core CPUs; 27,648 Nvidia Tesla V100 GPUs
- How fast is it? **122.3** pFLOPS
- (In computing, **floating point operations per second** (**FLOPS**, **flops** or **flop/s**) is a measure of computer performance, useful in fields of scientific computations that require floating-point calculations. It is a more accurate measure than measuring instructions per second.)

Task 2

- Just revise the following basic concepts:

- Architecture of microprocessor
- Types of Registers
- Types of memory